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Question 1

Solution:**A**

We have 8 timesteps t . So until $t=8$ we have to compute;

$$G_t = \sum_{k=0}^{t=7} R_{t+k+1}$$

This is without γ^k since it is 1 in this example. So there is no discounting.

For $t=0$, $St=A$

$$G_0 = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 + R_8$$

$$G_0 = 1 + 1 + 0 + 1 + 1 + 1 + 0 + 0$$

$$G_0 = 5$$

For $t=1$, $St=A$

$$G_1 = R_2 + R_3 + R_4 + R_5 + R_6 + R_7 + R_8$$

$$G_1 = 1 + 0 + 1 + 1 + 1 + 0 + 0$$

$$G_1 = 4$$

For $t=2$, $St=A$

$$G_2 = R_3 + R_4 + R_5 + R_6 + R_7 + R_8$$

$$G_2 = 0 + 1 + 1 + 1 + 0 + 0$$

$$G_2 = 3$$

For $t=3$, $St=B$

$$G_3 = R_4 + R_5 + R_6 + R_7 + R_8$$

$$G_3 = 1 + 1 + 1 + 0 + 0$$

$$G_3 = 3$$

For $t=4$, $St=A$

$$G_4 = R_5 + R_6 + R_7 + R_8$$

$$G_4 = 1 + 1 + 0 + 0$$

$$G_4 = 2$$

For $t=5$, $St=A$

$$G_5 = R_6 + R_7 + R_8$$

$$G_5 = 1 + 0 + 0$$

$$G_5 = 1$$

For $t=6$, $St=A$

$$G_6 = R_7 + R_8$$

$$G_6 = 0 + 0$$

$$G_6 = 1$$

For $t=7$, $St=B$

$$G_7 = R_8$$

$$G_7 = 0$$

First Visit

First-visit MC uses only the return at the first time the state is visited. First visit to A is at $t=0$, so $v(A)_{\text{first}} = G_0 = 5$ First visit to B is at $t=3$, so $v(B)_{\text{first}} = G_3 = 3$

Every Visit

Every-visit MC averages the returns from all times the state was visited. For A, visits and returns: $t=0,1,2,4,5,6$ with G values $[5,4,3,2,1,0]$. Sum them: $5+4+3+2+1+0 = 15$. Number of visits = 6. Average: $15 / 6 = 2.5$. So $v(A)_{\text{every}} = 2.5$ For B, visits and returns: $t=3,7$ with G values $[3,0]$. Sum: $3+0 = 3$. Number of visits = 2. Average: $3 / 2 = 1.5$. So $v(B)_{\text{every}} = 1.5$

B

So this time the sum becomes;

$$G_t = \sum_{k=0}^{t=7} \gamma^k R_{t+k+1}$$

for $t=0$, $St=A$

$$G_0 = R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5 + \gamma^5 R_6 + \gamma^6 R_7 + \gamma^7 R_8$$

Filling in the reward values;

$$G_0 = 1 + \gamma 1 + \gamma^2 0 + \gamma^3 1 + \gamma^4 1 + \gamma^5 1 + \gamma^6 0 + \gamma^7 0$$

Removing the zeros. The algebraic expression becomes:

$$G_0 = 1 + \gamma 1 + \gamma^3 1 + \gamma^4 1 + \gamma^5 1$$

Repeating the same steps for $t=3$, $St=B$

$$G_3 = R_4 + \gamma R_5 + \gamma^2 R_6 + \gamma^3 R_7 + \gamma^4 R_8$$

Filling in the reward values;

$$G_3 = 1 + \gamma 1 + \gamma^2 1 + \gamma^3 0 + \gamma^4 0$$

Removing the zeros. The algebraic expression becomes:

$$G_3 = 1 + \gamma 1 + \gamma^2 1$$

C

To Compute the first and every visit estimators with $\gamma = 0.95$ We have to recompute G_t for every timestep.

$$\begin{aligned}
 G_0 &= R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \gamma^4 R_5 + \gamma^5 R_6 + \gamma^6 R_7 + \gamma^7 R_8 \\
 &= 1 + 0.95(1) + 0.95^2(0) + 0.95^3(1) + 0.95^4(1) + 0.95^5(1) + 0.95^6(0) + 0.95^7(0) \\
 &= 1 + 0.95 + 0 + 0.857375 + 0.81450625 + 0.7737809375 \\
 &= 4.3956621875
 \end{aligned}$$

For $t = 1, S_t = A$

$$\begin{aligned}
 G_1 &= R_2 + \gamma R_3 + \gamma^2 R_4 + \gamma^3 R_5 + \gamma^4 R_6 + \gamma^5 R_7 + \gamma^6 R_8 \\
 &= 1 + 0.95(0) + 0.95^2(1) + 0.95^3(1) + 0.95^4(1) + 0.95^5(0) + 0.95^6(0) \\
 &= 1 + 0 + 0.9025 + 0.857375 + 0.81450625 \\
 &= 3.57438125
 \end{aligned}$$

For $t = 2, S_t = A$

$$\begin{aligned}
 G_2 &= R_3 + \gamma R_4 + \gamma^2 R_5 + \gamma^3 R_6 + \gamma^4 R_7 + \gamma^5 R_8 \\
 &= 0 + 0.95(1) + 0.95^2(1) + 0.95^3(1) + 0.95^4(0) + 0.95^5(0) \\
 &= 0 + 0.95 + 0.9025 + 0.857375 \\
 &= 2.709875
 \end{aligned}$$

For $t = 3, S_t = B$

$$\begin{aligned}
 G_3 &= R_4 + \gamma R_5 + \gamma^2 R_6 + \gamma^3 R_7 + \gamma^4 R_8 \\
 &= 1 + 0.95(1) + 0.95^2(1) + 0.95^3(0) + 0.95^4(0) \\
 &= 1 + 0.95 + 0.9025 \\
 &= 2.8525
 \end{aligned}$$

For $t = 4, S_t = A$

$$\begin{aligned}
 G_4 &= R_5 + \gamma R_6 + \gamma^2 R_7 + \gamma^3 R_8 \\
 &= 1 + 0.95(1) + 0.95^2(0) + 0.95^3(0) \\
 &= 1 + 0.95 \\
 &= 1.95
 \end{aligned}$$

For $t = 5, S_t = A$

$$\begin{aligned}
 G_5 &= R_6 + \gamma R_7 + \gamma^2 R_8 \\
 &= 1 + 0.95(0) + 0.95^2(0) \\
 &= 1
 \end{aligned}$$

For $t = 6, S_t = A$

$$\begin{aligned}
 G_6 &= R_7 + \gamma R_8 \\
 &= 0 + 0.95(0) \\
 &= 0
 \end{aligned}$$

For $t = 7, S_t = B$

$$\begin{aligned}
 G_7 &= R_8 \\
 &= 0
 \end{aligned}$$

First Visit

First-visit MC uses only the return at the first time the state is visited. First visit to A is at $t=0$, so $v(A)_{\text{first}} = G_0 = 4.395$ First visit to B is at $t=3$, so $v(B)_{\text{first}} = G_3 = 2.852$

Every Visit

Every-visit MC averages the returns from all times the state was visited For A, visits and returns: $t = 0, 1, 2, 4, 5, 6$ with $G_t = [4.3957, 3.5744, 2.7099, 1.95, 1.0, 0.0]$.

$$\text{Sum} = 4.3957 + 3.5744 + 2.7099 + 1.95 + 1.0 + 0 = 13.6299$$

$$\text{Number of visits} = 6$$

$$v(A)_{\text{every}} = \frac{13.6299}{6} = 2.2717$$

For B, visits and returns: $t = 3, 7$ with $G_t = [2.8525, 0.0]$.

$$\text{Sum} = 2.8525 + 0 = 2.8525$$

$$\text{Number of visits} = 2$$

$$v(B)_{\text{every}} = \frac{2.8525}{2} = 1.42625$$

Thus,

$$v(A)_{\text{every}} = 2.27, \quad v(B)_{\text{every}} = 1.43$$

Question 2**Solution:****A**

My proposed scenario: operating a 3D print farm where multiple printers run sequential prints. Where I already gained a lot of experience estimating how long it takes to complete a print job based on different models and printer settings. After upgrading some machines, I want to estimate how long the print sequence will take. With Monte Carlo updates, you need to wait until all the prints in the sequence finish before you can update your estimate of the total time. This means early errors won't be reflected until the entire sequence is complete. With Temporal-Difference (TD) updates, you can start updating your estimate as soon as each print finishes, using the observed time and your current predictions for the remaining prints. Because much of your previous experience (how long each print takes) is still relevant, TD can transfer this knowledge immediately to the new setup, even before you have completed many full printing sequences. So, in this scenario, TD updates would be better on average because they allow you to adapt quickly using partial information, rather than waiting for full episodes as Monte Carlo does.

B

In this case, TD methods are better because they bootstrap from your previous knowledge; your estimates of the time for each print step remain mostly valid. You can update predictions for the new setup incrementally as you observe partial results.

Monte Carlo methods, on the other hand, require complete print cycles to make any updates, which means they learn more slowly at first, especially when the environment has changed.

The same idea applies to the driving home example: if you move to a new building but still use the same highway, your knowledge about travel times on most of the route remains accurate. TD methods can leverage existing knowledge to adapt faster, while Monte Carlo must relearn from scratch by waiting until you reach home each time.

Question 3

Solution:**A****What happened in the first episode:**

The episode followed the path: $C \rightarrow B \rightarrow A \rightarrow \text{Terminal}$ (left)

The agent started at the initial state C , moved left to B , then to A , and finally terminated at the left terminal state, receiving a reward of 0.

Why only $V(A)$ was updated:

TD(0) updates states **during** the episode using the temporal difference error between consecutive states. The TD(0) update rule is:

$$V(s) \leftarrow V(s) + \alpha[R + \gamma V(s') - V(s)]$$

- When the agent transitioned from A to the left terminal state, it performed an update on $V(A)$ using the reward (0) and the value of the terminal state (0)
- States B and C , which were visited earlier in the episode, would have updated based on their successor states (A and B respectively), but since all non-terminal states were initialized to 0.5, their TD errors were zero, resulting in no change
- Only $V(A)$ had a non-zero TD error because it transitioned to a terminal state with value 0

B**Calculation:**

Given $\alpha = 0.1$ and assuming $V(A)$ was initially 0.5:

$$\begin{aligned} \text{TD error} &= R + \gamma V(\text{terminal}) - V(A) \\ &= 0 + 0 - 0.5 \\ &= -0.5 \end{aligned}$$

$$\begin{aligned}
 \Delta V(A) &= \alpha \times (\text{TD error}) \\
 &= 0.1 \times (-0.5) \\
 &= -0.05
 \end{aligned}$$

$$V(A)_{\text{new}} = 0.5 + (-0.05) = 0.45$$

Answer: $V(A)$ decreased by exactly **0.05** (from 0.5 to 0.45)

This matches the red line in Figure 3, which shows $V(A)$ at 0.45 after one episode.

Question 4

Solution:

A

Value Function for Policy 1: [1.32432149 1.86940636 2. 0.] Value Function for Policy 2: [0.74060774 1.77333233 2. 0.]

B

Value Function for Policy 1: [1.45979384 1.53492478 1.83684068 0.] Value Function for Policy 2: [1.17055466 -0.21085982 1.92984222 0.]

C

Value Function for Policy 1: [1.48043288 1.59102996 1.85020601 0.] Value Function for Policy 2: [1.2543447 0.01825648 1.93596137 0.]

D

Policy π_1 is superior to π_2 , as shown by consistently higher value function estimates across all three algorithms. From the initial state s_0 , we have $V^{\pi_1}(s_0) > V^{\pi_2}(s_0)$ in all cases: MC shows $1.324 > 0.741$, TD(0) shows $1.460 > 1.171$, and TD(λ) shows $1.480 > 1.254$. This consistency across different estimation methods confirms $\pi_1 > \pi_2$. Thus, π_1 makes better action choices that lead to higher rewards. Also, π_2 achieves negative or near-zero values in some states ($V^{\pi_2}(s_1) = -0.211$ in TD(0)), showing it sometimes follows suboptimal paths, whereas π_1 keeps positive values throughout all non-terminal states. So it consistently reaches more rewarding trajectories.